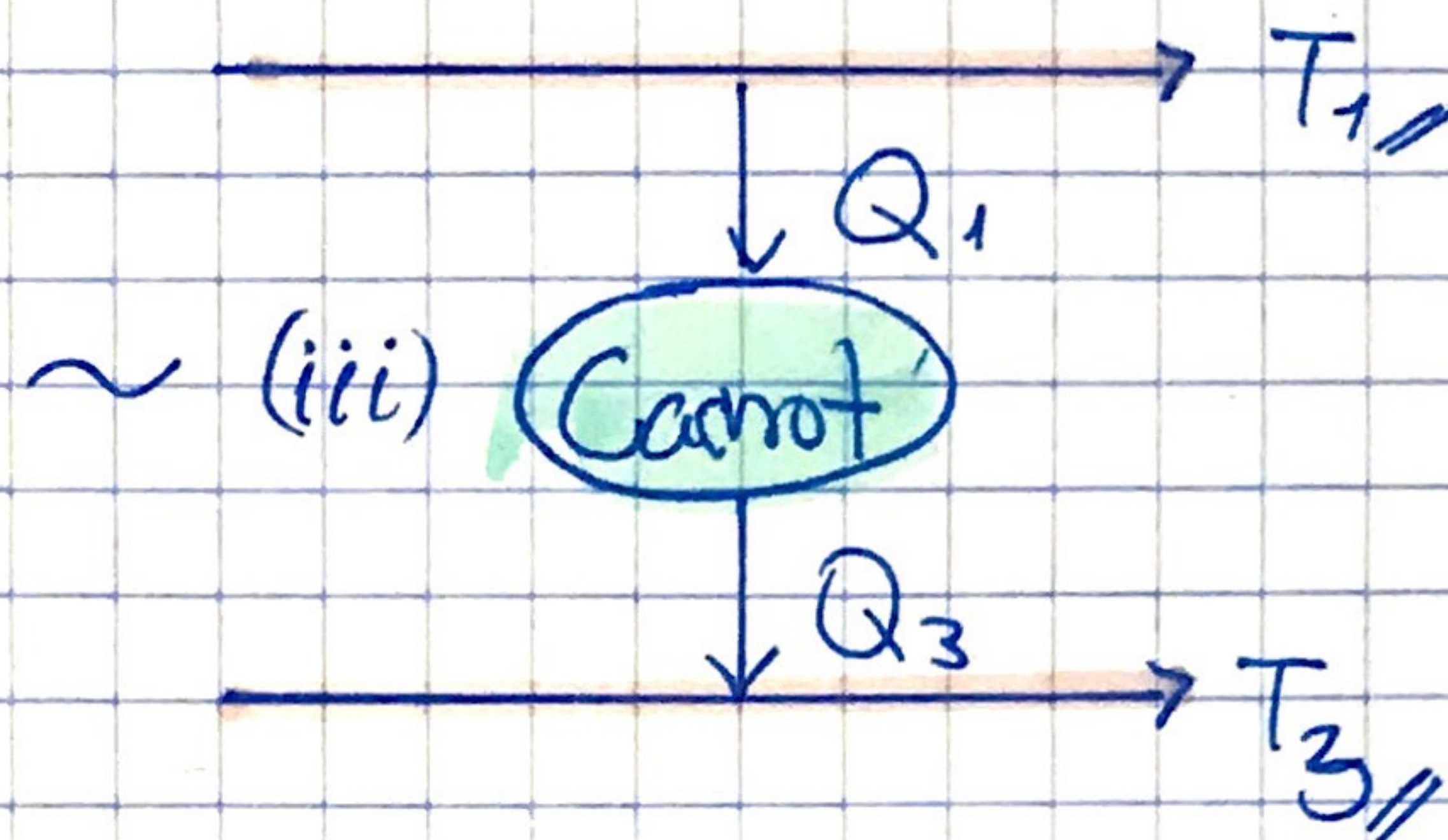
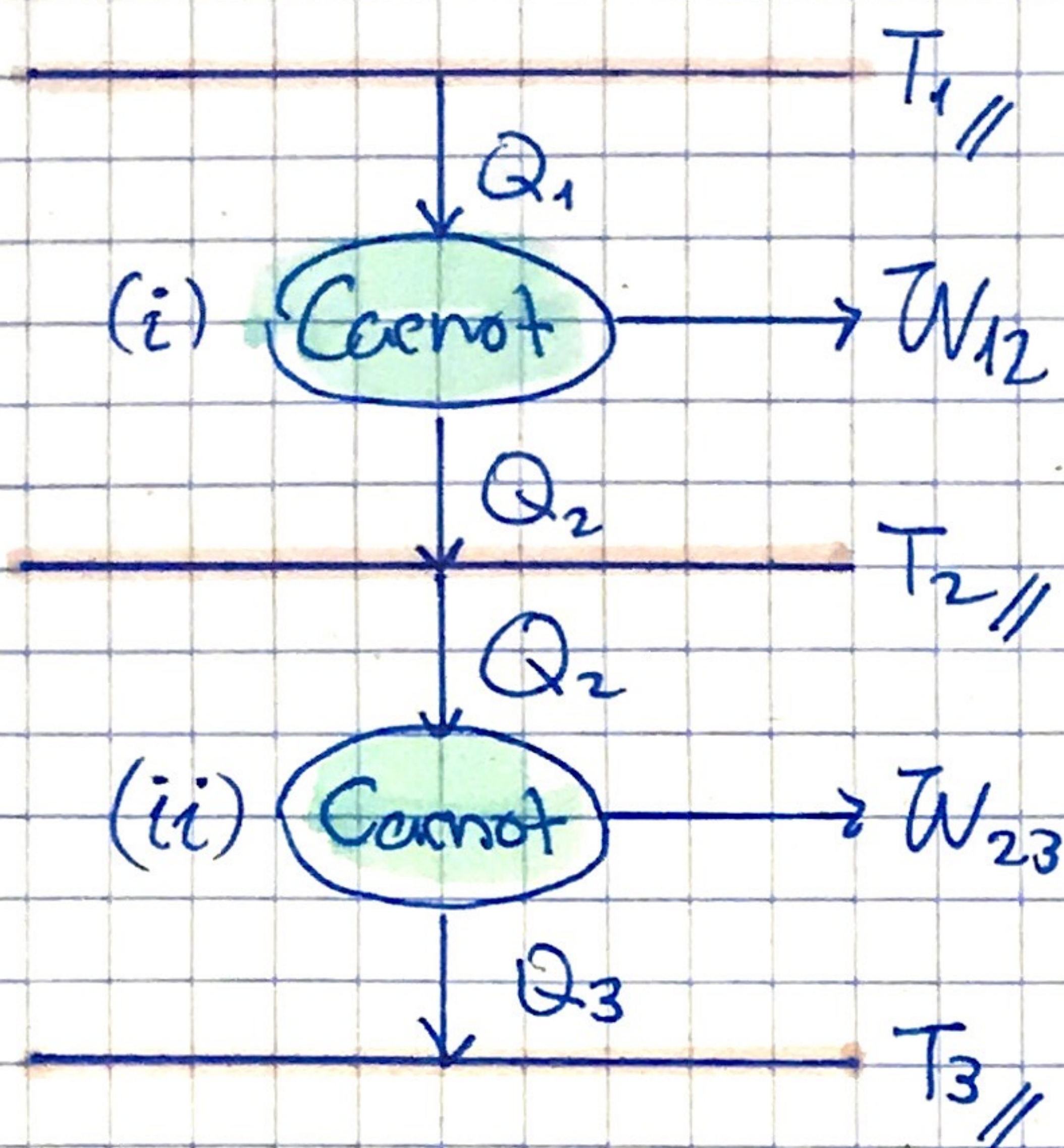




$$(i) \quad 0 = \Delta U = \Delta Q_{12} - W_{12} \Rightarrow Q_1 - Q_2 - W_{12} = 0$$

$$\left\{ \eta = \frac{W}{Q_{in}} \right\}$$

$$(1) \quad \begin{cases} Q_2 = Q_1 - W_{12} \\ = Q_1 - \cancel{Q_1 \eta(T_1, T_2)} - Q_1 \eta(T_1, T_2) \\ = Q_1 (1 - \eta(T_1, T_2)) \end{cases}$$

$$(ii) \quad 0 = \Delta U = \Delta Q_{23} - W_{23} \Rightarrow Q_2 - Q_3 - W_{23} = 0$$

$$(2) \quad \begin{cases} Q_3 = Q_2 - W_{23} \\ = Q_2 - Q_2 \eta(T_2, T_3) \\ = Q_2 (1 - \eta(T_2, T_3)) \end{cases}$$

$$\left\{ \begin{array}{l} (1) \rightarrow (2) \quad Q_3 = Q_1 (1 - \eta(T_1, T_2)) (1 - \eta(T_2, T_3)) \quad (3) \\ (iii) \quad Q_3 = Q_1 (1 - \eta(T_1, T_3)) \quad (4) \end{array} \right.$$

$$(5) \quad \boxed{1 - \eta(T_1, T_3) = (1 - \eta(T_1, T_2)) (1 - \eta(T_2, T_3))}$$